

RAUNAK SHRIVASTVA

AI / Deep Learning Engineer • GPU Acceleration (CUDA) • Systems & Web Architect

Jaipur, Rajasthan, India • raunakshrivastva@gmail.com • [linkedin.com/in/raunak-shrivastva](https://www.linkedin.com/in/raunak-shrivastva) • github.com/raunakshrivastva • x.com/raunak

PROFESSIONAL SUMMARY

Software Engineer specializing in scalable **Deep Learning architectures**, **high-performance CUDA kernel optimization**, and production-grade **full-stack reactive web systems**. Proven track record of applying parallel GPU compute primitives to accelerate model training and inferencing workloads, while pairing complex AI pipelines with hyper-responsive modern frontend architectures. Passionate about solving compute-bound engineering challenges across autonomous agent loops, hardware-accelerated systems, and low-latency interactive interfaces.

TECHNICAL SKILLS

AI & Deep Learning:	PyTorch, TensorFlow, Transformer Architectures, LLM Fine-Tuning (LoRA/QLoRA), Autonomous Agents, Computer Vision, Model Quantization, Hugging Face.
GPU Acceleration & HPC:	CUDA C/C++, Parallel Computing, GPU Kernel Optimization, Shared Memory Banking, Thread Coalescing, Memory Hierarchy, Profiling (Nsight Compute/Systems).
Frontend & Architecture:	TypeScript, React, Next.js, Vite, JavaScript (ES6+), Framer Motion, TailwindCSS, State Management (Zustand/Redux), WebGL Shaders, RESTful APIs.
Tools & Infrastructure:	Git, GitHub, Linux/Bash, Node.js, Webpack, Vite Build Tooling, Docker, Continuous Integration (CI/CD), Performance Benchmarking.

FEATURED ENGINEERING PROJECTS

Autonomous Agentic LLM Fine-Tuning & Reasoning Pipeline PyTorch , Transformers , QLoRA , Agentic Loops	2025 – 2026
- Engineered an autonomous multi-step reasoning framework using open-weight LLMs, attaining an 84% autonomous task completion rate on algorithmic benchmarks. - Optimized parameter-efficient fine-tuning (PEFT/QLoRA) across custom domain datasets, cutting VRAM overhead by 62% within a 0.8% perplexity delta. - Implemented dynamic context-caching and vector retrieval, slashing token generation latency by 3.4x during autonomous trajectory planning.	
High-Throughput CUDA GPU Acceleration & Matrix Engine CUDA C++ , C++20 , Nsight	2025
- Architected a high-performance parallel compute engine in CUDA C++, reaching 91% theoretical peak memory bandwidth on NVIDIA hardware for dense GEMM ops. - Eliminated shared memory bank conflicts and uncoalesced memory transactions, yielding an immediate 4.8x kernel speedup over standard implementations. - Profiled compute hotspots using NVIDIA Nsight Compute, tuning block dimensions and register allocation to drive SM occupancy above 88%.	
Neural Interface & Interactive Web Architecture ("Wayne's World") Next.js , TypeScript , Framer Motion , WebGL	2025 – 2026
- Designed and developed a cinematic neural interface UI platform featuring custom WebGL shaders and real-time cursor-tracking spotlights locked at 60 FPS. - Built a resilient micro-interaction engine with Framer Motion, reducing layout thrashing and cumulative layout shift (CLS) to 0.00 across desktop and mobile. - Architected modular client-side state and optimized bundle distribution, delivering sub-800ms First Contentful Paint (FCP) and cutting bundle size by 38%.	
Distributed Deep Learning Vision & Inference Pipeline PyTorch , TensorRT , Linux , REST API	2024 – 2025
- Deployed high-efficiency deep convolutional and attention vision models, scaling batch inference throughput past 1,200 requests/sec via asynchronous pipeline streaming. - Quantized FP32 checkpoints to INT8/FP16 precision via TensorRT, shrinking disk footprint by 4x while slashing end-to-end response times by 52%.	

CERTIFICATIONS & ACCREDITATIONS

NVIDIA / Industry Accredited Certification in CUDA Acceleration & Parallel Computing GPU architecture, CUDA thread hierarchy, memory optimization, and multi-threaded parallelization.	Certified
Deep Learning Specialization Certification (DeepLearning.AI) Neural Networks, Backpropagation calculus, CNNs, Sequence Models, Transformers, and Hyperparameter Optimization.	Certified

EDUCATION

JECRC University — Bachelor of Technology (B.Tech) in Computer Science & Engineering Coursework: High-Performance Computing, Machine Learning, Systems Programming, Data Structures & Algorithms.	Class of 2026 Jaipur, India
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